



Search for novel biomarkers of febrile neutropenia in hematological patients: an overview

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Received: 15 September 2025 / Accepted: 1 December 2025
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Abstract

A single-center prospective study was conducted at a tertiary care hospital in Kuopio Finland over the past two decades, searching for novel biomarker candidates to predict complicated course of febrile neutropenia (FN) in hematological patients. In Dec 2006–Dec 2015, 85 patients with acute myeloid leukemia and 179 autologous hematopoietic stem cell transplant recipients were included. More than 20 biomarkers were evaluated in subsequent patient cohorts and compared to C-reactive protein (CRP) and procalcitonin (PCT). The samples were taken on d0-d2 from the beginning of FN. Here, we provide an overview of the results of these studies. Most biomarkers evaluated did not provide additional prognostic benefit compared to CRP or PCT. The most promising biomarkers warranting further studies include soluble cluster of differentiation 14, interleukin-1-receptor antagonist, interleukin-10, tissue inhibitor of matrix metalloproteinase-1, and caspase-cleaved cytokeratin-18. Several biomarkers evaluated in this series of studies are hampered by impaired production by neutrophils/leukocytes (matrix metalloproteinase-8, cell-free plasma DNA and soluble urokinase plasminogen activator receptor) or platelets (vascular endothelial growth factor). For now, these novel biomarkers cannot substitute the conventional markers nor provide additional benefit in routine use. Optimally, the most promising novel markers should be evaluated in prospective multicenter studies with higher patient numbers and several endpoints to more reliably evaluate their performance. At present, close patient monitoring is imperative in hematological patients with FN. The additional information provided by novel FN biomarkers may be useful, but they should be evaluated combined with daily evaluation of sepsis scoring systems.

Keywords Febrile neutropenia · Complicated course · Prognosis · Biomarker · Acute myeloid leukemia · Autologous hematopoietic cell transplantation

Introduction

Febrile neutropenia (FN) is a common consequence of intensive chemotherapy in hematological patients. It is associated with severe complications such as bacteremia, sepsis, need for intensive care unit (ICU) admission, and

even death. In addition to neutropenia *per se*, there are other factors predisposing hematological patients treated with intensive chemotherapy to severe infections, such as oral and intestinal mucositis, long-term vascular catheters, and immunosuppression due to disease and/or its treatment. In these circumstances, signs of infection other than fever may be absent or minimal. Although infection is the most common cause of fever in severely neutropenic patients, it may be also caused, e.g., by mucosal damage or medication. Therefore, biomarkers of FN have long been sought to improve diagnostic and prognostic assessment in patients with FN.

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Why do we need sepsis biomarkers?

Diagnosis of a potentially life-threatening infection remains challenging. In neutropenic immunocompromised hematological patients receiving intensive chemotherapy, symptoms and signs of infection may be subtle and minimal. These patients may also have other causes of fever, including viral infections, mucosal damage, or medications, notably cytarabine. Obtaining blood culture results with conventional methods takes too long considering the patient's potentially unstable condition. Further, blood cultures are frequently negative in patients with FN. Therefore, **diagnostic** use of biomarkers is important in this patient population. Although all biomarkers are non-specific, they may raise clinical suspicion of sepsis.

Most commonly biomarkers have been used for **prognostic** purposes. This applies especially to patients treated in emergency department or at hematology ward. Biomarkers may offer useful information about patient's follow-up requirements and the need to enhance treatment intensity. This is also true for patients treated at ICU although several physiological measurement results are available in real time.

Biomarkers may be useful in the evaluation of **treatment efficacy**. Biomarkers may help the clinician to assess the patient's current condition and treatment response. Biomarker monitoring may also enable de-escalation of antimicrobial therapy.

Novel biomarkers may also provide clues to the underlying **pathophysiological mechanisms**, thus increasing our understanding of the complex and multifaceted background of sepsis syndrome. Table 2 shows examples of biomarkers grouped by proposed pathophysiological background.

Epidemiological background

Epidemiological studies were crucial for the start of this single center prospective study series. In a retrospective study on patients with acute myeloid leukemia (AML) aged < 70 years treated at Kuopio University Hospital (KUH) in 1996–2005 ($n = 84$) [1], FN was observed in 97% of the patients (280 FN periods) following induction or consolidation courses. Blood cultures were positive in 59% of FN periods, and severe sepsis was diagnosed in 13% of FN periods according to the 2003 definition by International Sepsis Definitions Conference [2]. Thirteen years later, sepsis was defined according to the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3) as a life-threatening organ dysfunction caused by dysregulated host response to infection [3]. The Quick Sequential Organ Failure Assessment (qSOFA) score was defined by retrospectively assessing organ dysfunction: (a) respiratory rate

$\geq 22/\text{min}$, (b) altered mental status (Glasgow Coma Scale < 15), and (c) systolic blood pressure ≤ 100 mm Hg, one point for each dysfunction, maximum 3 points. Septic shock was regarded as a subset of sepsis, in which underlying circulatory, cellular, and metabolic abnormalities are profound enough to substantially increase mortality [4]. Gram-negative bacteremia was more common in patients with severe sepsis than in patients without severe sepsis (40 vs. 23%). 5% of FN periods led to ICU admission and eight patients (9.5%) died due to sepsis. A nationwide prospective study by the Finnish Leukemia Group included a total of 357 AML patients with 977 treatment cycles. Bacteremia was observed in 51% of these episodes, and 5.6% of patients died due to infectious complications [5].

A retrospective evaluation of 319 autologous hematopoietic cell transplant (AHCT) recipients was conducted at KUH in 1995–2005. FN was observed in 83% and blood cultures were positive in 25% of patients. Patients transplanted for non-Hodgkin lymphoma (NHL) were three times more likely to have severe sepsis (9% vs. 3%) compared to other patients [6]. Altogether 3.4% of the patients died due to infectious complications, especially *Pseudomonas aeruginosa* bacteremia. Our prospective multicenter GOA study (Graft and Outcome in Autologous transplantation) included 147 multiple myeloma (MM) patients receiving AHCT at four university hospitals in Finland in 2012–2016. At the same time, 136 NHL patients were also included in the study. In MM patients, FN was observed in 79% and bacteremia in 18% of the patients after AHCT. There were no early deaths (< 100 d) or need for ICU admissions [7]. In NHL patients, the incidence of post-transplant FN was 88%, bacteremia 17% and ICU admission 4%. Two patients (1.5%) had transplant-related mortality within 100 days from AHCT [8].

In our prospective biomarker studies, we focused on patients at substantial risk of FN-associated complications, i.e. AML patients receiving intensive chemotherapy and AHCT recipients. The initial upper age limit of 70 years was subsequently removed when more recently selected elderly patients were treated with standard AML induction and consolidation courses. Also selected elderly patients up to 75 years of age were accepted to proceed to AHCT.

Study protocol

Methods

The study protocol was accepted by the Ethical Board of North Savo Hospital District in 2006 (100/2006) with several subsequent amendments. Informed consent was obtained from all individual participants. Between December 1,

2006, and December 31, 2015, more than 300 adult patients (> 16 years of age) with AML or AHCT were recruited in this single center prospective HEMATULEHDUS study. Three successive study cohorts were formed including altogether 264 patients (AHCT 179, AML 85) who fulfilled the entry criteria for FN (oral temperature > 38 °C at least for an hour and neutrophils < $0.5 \times 10^9/L$ or WBC count < $1.0 \times 10^9/L$ with predicted decrease to < $0.5 \times 10^9/L$) (Table 1). In principle, only the first FN period was included in both patient groups. At the onset of fever (d0), blood cultures were taken immediately from cubital veins and broad-spectrum antibiotics were started after the samples. Most

Table 1 Study cohorts of hematological patients with febrile neutropenia during HEMATULEHDUS study at Kuopio University Hospital between December 1, 2006, and December 31, 2015

	Cohort 1	Cohort 2	Cohort 3
Time period	December 1, 2006–November 30, 2009	December 1, 2009–November 30, 2012	December 1, 2012–December 31, 2015
Patients with FN (N)	100	86	78
AML (N)	32	23	30
AHCT (N)	68	63	48
Bacteremia (%)	19%	21%	28%
ICU admission N (%)	5 (5%)	6 (7%)	1 (1%)
In-hospital mortality N (%)	3 (3%)	4 (3.5%)	0 (0%)
<i>Biomarkers evaluated:</i>			
	Antithrombin	ADMA	Alpha1-AGP
	Copeptin	Caspase-cleaved cytokeratin-18	Citrulline
	Fibrinogen	Cell-free plasma DNA	I-FABG
	IL-10	IL-18	MMP-8
	IL-6	IL-1Ra	ZO-1
	NT-proBNP	Metabolomics	
	PCT	MMP-10	
	Plasma lactate	sCD14	
	PTX 3	TIMP-1	
	Serum cortisol		
	suPAR		
	VEGF		

ADMA, asymmetric dimethylarginine; AHCT, autologous hematopoietic cell transplantation; alpha1-AGP, alpha-1 acid glycoprotein (orosomucoid); AML, acute myeloid leukemia; BNP, brain-type natriuretic peptide; FN, febrile neutropenia; ICU, intensive care unit; I-FABG, intestinal fatty acid binding globulin; IL-10, interleukin-10; IL-18, interleukin-18; IL-1Ra, interleukin-1 receptor antagonist; IL-6, interleukin-6; MMP-10, matrix metalloproteinase-10; MMP-8, matrix metalloproteinase-8; NT-proBNP, amino-terminal pro-brain natriuretic peptide; PCT, procalcitonin; PTX-3, pentraxin-3; sCD14, soluble cluster of differentiation 14; suPAR, soluble urokinase plasminogen activator receptor; TIMP-1, tissue inhibitor of matrix metalloproteinase-1; VEGF, vascular endothelial growth factor; ZO-1, zonulin 1

NHL patients undergoing AHCT received antibiotic prophylaxis with ciprofloxacin. As to AML patients, broad-spectrum antibiotic therapy (third-generation cephalosporin and tobramycin or piperacillin-tazobactam and tobramycin) was started after blood cultures. Vancomycin was considered only if there was evidence of infection in the central venous catheter or if microbiological results showed gram-positive bacteria. Antimicrobial therapy was re-evaluated based on microbiological, radiological and clinical findings. New blood cultures were obtained if fever persisted for 3–5 days. Previously, oral fluconazole was started at the onset of neutropenia and posaconazole was started as a prophylactic antifungal from 2009 onwards in AML patients. If fever persisted, caspofungin was started empirically to prevent possible fungal infection. Samples for other biomarker studies were taken simultaneously from cubital veins followed by sampling on the next two mornings (d1–d2). Initially, samples were collected over five days, but later the sampling of study markers was reduced to three samples (d0–d2) because we wanted to specifically assess early biomarkers of FN. Novel markers were compared to CRP measured at the same time points, and later also to PCT. The study endpoints were (1) positive blood culture, (2) fulfillment of criteria for severe sepsis [2], (3) ICU admission, and (4) death. Laboratory methods are described in original articles.

Supportive care

All FN patients were treated at the adult hematology ward with close follow-up of vital functions. The patients were cared for in single rooms by experienced and well-trained nursing personnel. Vital signs such as body temperature, blood pressure, heart rate, respiration frequency, and peripheral blood oxygen saturation were regularly assessed after the onset of FN. Fluid intake (oral and infused) and urine output were registered every 12 h for the first three days of FN. Oxygen supplementation was given if needed to maintain oxygen saturation > 90%. In case of clinical deterioration, ICU admission or step-up step-down unit admission was actively considered.

Statistical methods

In summary, data analyses were conducted with SPSS version 19–23 for Windows (SPSS, Chicago, IL, USA). Categorical variables were given as absolute counts or frequencies. Correlations between continuous variables were analyzed by Spearman's correlation test. Chi-squared test was used to evaluate the association between categorical variables. Also repeated measures analysis of variance was used. Variables under interest were reported as medians and interquartile ranges (or means with standard deviations)

according to the presence of complications of FN. The non-parametric Mann-Whitney *U*-test was used to evaluate the differences of continuous variables between the groups. Logistic regression analysis was used in case of a binary outcome (e.g., yes/no, 0/1) based on one or more explanatory variables. Receiver operating characteristic curve (ROC) analyses were carried out to evaluate the prognostic ability of markers, providing values for the area under the curve (AUC) with 95% confidence intervals (CI). The significance of differences between the AUCs was tested when needed according to Hanley and McNeil [9]. Sensitivity, specificity, positive and negative predictive value, positive likelihood ratios (LR+), negative likelihood ratios (LR-) and diagnostic odds ratios (DOR) were used as diagnostic parameters. Youden's indices (sensitivity + specificity - 1) were used when defining the optimal cut-off values for markers on each study day from day 0 to day 2 or from d0 to d4. The confidence intervals for Youden's index were calculated according to Shan et al. [10]. Weighted Youden's indices were also used to determine alternative cut-off values [11]. A *P* value of less than 0.05 was considered significant.

In general, the predictive ability of the novel markers were compared with the markers already in established use (CRP, PCT). The endpoints used were positive blood culture, sepsis, severe sepsis, septic shock, intensive care admission and death. The effect of underlying disease, age, gender, hematological disease, and drug therapy used was assessed and taken into account in the analyses where needed.

Main results of biomarker studies

The main numerical results have been collected in a single supplementary table (Table S1).

Serum vascular endothelial growth factor (VEGF)

VEGF has physiological roles both in the maintenance of vascular permeability and in angiogenesis [12]. Earlier studies on VEGF have been mainly performed in sepsis patients treated at ICU [13, 14]. We evaluated VEGF-A isoforms VEGF₁₂₁ and VEGF₁₆₅ among 42 hematological patients with FN treated at the hematology ward. Patients with severe sepsis had significantly higher VEGF values than other patients on d0 and d1 [15]. However, all these values were significantly lower than the reference values, most probably due to severe thrombocytopenia during the sampling period.

VEGF may be an earlier indicator of severe sepsis than CRP [15], apparently due to its rapid liberation from endothelial cells and platelets. Platelet counts influence

serum VEGF levels, making interpretation difficult in the context of thrombocytopenia. There is an obvious need to investigate serum VEGF concentrations in thrombocytopenic patients without signs of infection (e.g., patients with immune thrombocytopenia) to evaluate reference values for patients with severe thrombocytopenia.

N-terminal brain-type natriuretic peptide (NT-proBNP)

NT-proBNP belongs to the family of natriuretic peptides and acts as a cardiac hormone, as does also atrial natriuretic peptide. Cardiomyocytes secrete it in response to increased myocardial wall stress. It has both natriuretic and vasodilatory properties that tend to maintain cardiovascular homeostasis [16]. We evaluated NT-proBNP in 70 hematological patients with FN. NT-proBNP values did not significantly differ between patients with and without severe sepsis on d0-d4 [17]. The values obtained were much lower than those of ICU patients in previous sepsis studies [18, 19]. We observed that patients with a previous cardiovascular history had elevated NT-proBNP values on d0-d4. Of note, 90% of the patients had received anthracyclines (doxorubicin or idarubicin), which may affect NT-proBNP values.

For these reasons, NT-proBNP does not appear to have any benefit or added value from a diagnostic or prognostic perspective in hematological patients with FN treated at hematology ward. Hypothetically, it might be relevant in optimizing supportive care in FN patients with pre-existing cardiac disease.

Plasma fibrinogen and antithrombin

In sepsis, the immune response to infection activates the coagulation cascade resulting in microcirculatory clot formation. The laboratory alterations include decreased fibrinogen and decreased antithrombin levels. Antithrombin serum level is usually decreased in sepsis, where there is disseminated intravascular coagulopathy (DIC). In Japan, antithrombin, widely used in clinical practice, is indicated for use in patients with DIC with an antithrombin activity $\leq$ 70% or lower [20].

We observed that antithrombin activity decreased among 69 patients with FN and the decrease was more pronounced in patients with severe sepsis on d3 and d4 [21]. However, the decrease was modest, and only few patients had activity below 60%. On the contrary to our expectations, plasma fibrinogen values were higher in patients with severe sepsis. On d0, plasma fibrinogen level > 4 mg/L had a positive predictive value of only 17% for severe sepsis but the negative predictive value was 100%. Plasma fibrinogen level correlated with CRP values on d0-d4.

It appears that plasma fibrinogen is not useful in the evaluation of hematological patients with FN treated at hematology ward when added to CRP.

Plasma lactate

Tissue hypoperfusion in sepsis can lead to insufficient oxygen supply, anaerobic metabolism and excessive lactate production. Lactate measurements are commonly used in the emergency room for patients suspected of being in critical condition and at ICU to monitor the adequacy of supportive care.

We evaluated serial plasma lactate levels among 70 hematological patients (94 FN periods) on d0-d3 from the onset of FN. On d0, the plasma lactate level was above the upper limit of normal in only four periods (5%) and did not differ between patients with severe sepsis and other patients in the early days of FN [22]. Routine serial lactate measurements are not useful in the evaluation of FN patients treated at hematology ward. However, in case of clinical deterioration (poor oxygen saturation, hypotension, declining diuresis, or decreased consciousness), lactate measurements are valuable together with consideration of transfer to ICU or to step-up step-down unit if available.

Serum cortisol

Despite its physiological importance, serum cortisol levels have not been systematically measured in hematological patients treated at ward for FN. We evaluated serum cortisol levels from the onset of FN among 69 hematological patients with FN [23]. Corticosteroids have long been used in the treatment of such patients with septic shock, who respond poorly to vasoactive agents [24, 25]. Serum cortisol levels correlated with CRP and PCT values on d0-d2 in patients without severe sepsis, but not in those with severe sepsis possibly suggesting an attenuated cortisol response in these patients. This implies that there may be subclinical adrenal insufficiency in these patients, which may call for further studies.

Serum cortisol measurements do not appear to offer additional benefit compared to more conventional biomarkers in hematological patients treated at ward.

Plasma copeptin

Copeptin is a 39 amino acid peptide. It is released in equimolar ratio to arginine vasopressin (AVP) to maintain hemodynamic balance by regulating volume status and hypothalamic-pituitary-adrenal axis. It has been found to be a prognostic marker in patients with septic shock [26].

We evaluated copeptin in daily measurements among hematological patients with FN. Copeptin levels did not correlate with complicated course of FN, but the lack of increase from d0 to d1 was associated with subsequent bacteremia and septic shock [27]. It has been speculated that impaired regulation of AVP release would hamper the use of copeptin as an indicator of the severity of septic state and would explain the discrepancies between studies [28]. Although the current results on the use of copeptin for diagnostic and prognostic purposes in FN are limited to our pilot study, copeptin does not appear to be useful in the evaluation of FN patients treated in the hematology ward.

Pentraxin-3 (PTX-3)

PTX-3 is a long-acting member of pentraxin an acute phase protein whose plasma levels are rapidly increased in various inflammatory states, including sepsis [29]. It predicts severe disease in febrile patients in emergency department [30]. In patients with sepsis, high PTX-3 levels have been associated with elevated risk of severe sepsis and mortality [31, 32].

As there are no previous studies of PTX-3 in hematological patients with FN, we evaluated PTX-3 among 100 patients with FN. The rise in PTX-3 was faster than the rise in CRP [33]. PTX-3 levels on d0 correlated with the development of septic shock. Non-survivors had constantly high PTX-3 levels. The cut-off levels predicting complicated course was much lower in AML patients compared to AHCT recipients with lymphoma [34]. This observation remains without an explicit explanation and needs to be confirmed. More clinical studies are needed to justify clinical use of PTX-3.

Procalcitonin (PCT)

Procalcitonin has been extensively studied for two decades in sepsis patients. It was approved by the Food and Drug Administration in 2020 to predict the development of complicated course of sepsis. It can also be used in de-escalation strategies of antimicrobial therapy, although the relatively long half-life (about 24 h) is somewhat problematic here. It is currently perhaps the most widely studied biomarker of FN in hematological patients.

In our prospective study, PCT was initially evaluated from stored samples with a semi-quantitative method. Ninety FN periods in 66 patients were evaluated. Bacteremia was observed in 21 episodes (23%), and half of them were gram-negative [35]. PCT level on d1-d2 was associated with bacteremia and gram-negative bacteremia, but not with the development of severe sepsis. On d1 PCT values >

0.5 µg/L had a sensitivity of 70% and specificity of 77% to predict gram-negative bacteremia.

In a subsequent study, quantitative PCT was evaluated in a cohort of 100 FN patients [36]. The area under the curve showed statistical significance for PCT to predict complicated course of FN already on d1 after the onset of FN. With a cut-off point of 0.13 microg/L on d0 or d1, sensitivity was 0.95, specificity 0.53 and negative predictive value 0.98 but positive predictive value was only 0.53.

Interleukin-6 (IL-6)

IL-6 is a proinflammatory cytokine that closely reflects the severity of the inflammatory response. Elevated IL-6 levels are associated with increased mortality and risk of multiple organ dysfunction in patients with critical illness [37]. After PCT and CRP, IL-6 is perhaps the most widely used biomarker of sepsis. Automatized laboratory methods are available in clinical laboratories to measure IL-6 24/7.

We evaluated IL-6 in the cohort of hematological FN patients in comparison with CRP, PCT and IL-10. The increase in IL-6 values was early and it normalized quickly. Considering Youden's indices, the increase in IL-6 was a good indicator of the gram-negative bacteremia and septic shock. With a cut-off value of 330 ng/L, sensitivity was 0.60 and specificity 0.92 [36]. On d1, the negative predictive value was as high as 0.95.

Interleukin-10 (IL-10)

IL-10 is an anti-inflammatory cytokine produced by macrophages and lymphatic cells but also by endothelial cells and neutrophils. IL-10 acts as an anti-inflammatory cytokine, inhibiting the production of several proinflammatory mediators, including tumor necrosis factor alpha (TNF-α).

In hematological FN patients after intensive chemotherapy, we observed the highest IL-10 values on d1. The concentrations were higher in patients with a complicated course of FN [36]. In comparison with other markers, IL-10 reached the highest Youden's index level followed by IL-6, PCT and CRP. In our study, with a cut-off value of 37 ng/L for prediction of gram-negative bacteremia or septic shock, the sensitivity was 1.00, specificity 0.79 and negative predictive value 1.00. In combination with PCT, IL-10 showed better prognostic power for complicated course of FN [36].

According to our study, IL-10 measurement would provide additional information about the early stage of FN in hematological patients. Unfortunately, at present there are no methods available for 24/7 use in clinical laboratory.

Soluble form of urokinase activator receptor (suPAR)

Plasma suPAR levels increase with immune activation. Previous studies indicate that a high plasma concentration of suPAR predicts poor outcome in infectious diseases, and a high suPAR level has found to be an independent risk factor for case fatality [38, 39].

We evaluated suPAR in 99 hematological patients with FN. suPAR values were higher on d0 and d1 in patients with gram-negative bacteremia and in those who developed a septic shock [40]. When compared to PCT measured at the same time points, suPAR was inferior to predict complicated course of FN.

In patients with FN, suPAR does not appear to be an ideal early biomarker as the main source of its production are neutrophils [41].

Cell-free plasma DNA (cf-DNA)

In cancer patients, cf-DNA originates from apoptotic and necrotic cancer cells or blood cells, mainly neutrophils and lymphocytes. In ICU-treated sepsis patients, cf-DNA levels have been found to predict case fatality [42, 43].

We prospectively evaluated cf-DNA levels in hematological patients with FN. We found no difference in cf-DNA levels between patients with or without a complicated course of FN [44]. In subgroups according to hematological malignancy, patients with acute myeloid leukemia (AML) had lower cfDNA levels than patients with lymphoma. Among AML patients, d0 cfDNA/leukocyte ratio and among lymphoma patients d0 cfDNA was associated with subsequent development of sepsis or septic shock.

Because the level of cf-DNA is influenced by the underlying disease and leukocyte count, it is difficult to interpret and thus suboptimal as a marker in severely leukopenic and neutropenic hematological FN patients.

Asymmetric dimethylarginine (ADMA)

ADMA is an inhibitor of endogenous nitric oxide, which is considered to play a key role in the development of sepsis-associated vasodilatation and hypotension. Elevated ADMA levels have been associated with organ damage and poor prognosis in sepsis patients treated at ICU [45, 46].

We evaluated ADMA among 87 patients with FN. ADMA levels did not change from d0 to d2. Organ dysfunction was rare in our series (7%) and only seven patients were treated at ICU. ADMA was not able to distinguish hematological patients with a complicated course of FN (bacteremia, severe sepsis or ICU admission) from those without

complications. We believe that ADMA is not useful in FN patients treated at the hematology ward [47].

Non-targeted metabolomics

Non-targeted metabolomics has been mainly investigated in sepsis patients treated at ICU [48–50]. Only few studies are currently available in hematological patients with FN.

We studied metabolomics among 85 patients with FN from samples collected on d0–d1. Altogether 90 molecular features differed between complicated (bacteremia, severe sepsis, ICU admission, death) and non-complicated course of FN. The most striking differences were observed in citrulline, androgen hormone and phosphatidylethanolamine [51].

In a study among hematological neutropenic patients from Germany [52], metabolomics was found to be superior to conventional biomarkers such as CRP, PCT and IL-6 at the onset of fever in patients with a verified infection. Threonine, tryptophan and lysophosphatidylcholine levels predicted FN resolution. Another study, including 14 bacteremic and 25 non-bacteremic patients, reported that a 5-predictor metabolomic model had an area under the receiver operating characteristic curve of 0.991 with 100% sensitivity, and 96% specificity in identifying bacteremia. Pregnenolone showed increased levels in patients with bacteremia [53].

Although currently based on only a few studies, non-targeted metabolomics may soon prove to be a useful adjunct in the evaluation of patients with FN. It may provide insights into the complex pathophysiology of FN and sepsis. It may also generate clues about potential therapeutic targets in these patients.

Soluble CD14 (sCD14)

CD14 is a cell surface glycoprotein expressed in monocytes, macrophages, neutrophils and B lymphocytes. It binds to various ligands on the surfaces of bacteria and is released to circulation as soluble CD14 (sCD14). Although sCD14 has been extensively studied, only a few studies have evaluated sCD14 in hematological patients with FN [54].

We evaluated sCD14 in hematological patients with FN. The samples were collected on d0–d2. The predictive power for the complicated course of FN was compared to that of PCT and CRP. The level of sCD14 was not associated with blood culture positivity or development of severe sepsis. However, on d1, the sCD14 level predicted the development of septic shock [55]. It appears that sCD14 might be an early marker of poor prognosis in hematological patients with FN. Further studies in this patient population are warranted.

Combining sCD14 determination with PCT assay or with clinical sepsis scoring systems deserves further studies.

Interleukin-1 receptor antagonist (IL-1Ra)

IL-1Ra is a cytokine that binds to IL-1 receptor and inhibits signaling. It antagonizes the effects of interleukin-1 alpha and beta. A slightly modified IL-1Ra (anakinra) has been used for treatment of rheumatoid arthritis [56], Still's disease [57] and more recently cytokine storm induced by chimeric antigen receptor T-cell therapy (CAR-T) [58]. Only a few small studies are available on IL-1Ra in patients with FN [59, 60].

We investigated IL-1Ra compared with CRP and PCT in 86 hematological patients with FN. IL-1Ra was significantly elevated on d0 in patients who later developed bacteremia. In hematological patients, IL-1Ra has a comparable ability with CRP and PCT to predict severe sepsis at the early phase of FN [61]. IL-1Ra appears to be an early marker of complicated course of FN. Thus, further studies of IL-1Ra in hematological patients with FN are warranted.

Matrix metalloproteinase-10 (MMP-10)

MMP-10 is a member of metalloproteinase family. It is associated with chemotherapy-induced mucosal damage or mucositis [62]. MMP-10 was evaluated among 91 hematological patients with FN [63]. Eight patients developed severe sepsis and three of them died.

Serum concentrations of MMP-10 were elevated in patients with severe sepsis. Patients with severe sepsis had significantly higher MMP-10 levels on d1 and d2 than other patients. ROC curve analysis suggested that MMP-10 could serve as a valid early indicator of sepsis severity in patients with FN.

Tissue inhibitor of matrix metalloproteinase-1 (TIMP-1)

TIMP-1 is an inhibitor of matrix metalloproteinases, but it also increases cellular proliferation and prevents apoptotic cell death [64]. High levels of TIMP-1 have been associated with high mortality in non-neutropenic sepsis patients [65, 66], but data on subgroups of neutropenic hematological patients are lacking. Compared to patients with non-severe sepsis, FN patients with severe sepsis had higher TIMP-1 levels from d0 to d2 [63]. According to ROC curve analysis, the area under the curve values on d0 were higher for TIMP-1 than for CRP or PCT.

TIMP-1 appears to be a promising prognostic marker for FN in hematological patients. Further studies are needed to determine whether it is also a useful marker of mucosal

damage, as mucositis and FN usually coexist in hematological patients after intensive chemotherapy.

Interleukin-18 (IL-18)

IL-18 is a pro-inflammatory cytokine produced by macrophages and dendritic cells. Elevated levels have been measured in patients with sepsis, and prognostic value has been demonstrated in patients with severe sepsis or septic shock [67, 68]. IL-18 has not been previously studied in hematological patients with FN.

We evaluated IL-18 levels among 92 hematological patients with FN. Serum IL-18 level did not correlate with bacteremia or development of septic shock, but it correlated inversely with blood leukocyte count [69]. For unknown reasons, AHCT recipients with lymphoma had significantly higher IL-18 values than AML patients. IL-18 is not a suitable biomarker for hematological patients with FN due to its poor diagnostic performance and differences in measured levels according to underlying disease.

Caspase-cleaved cytokeratin 18 (ccCK-18)

Cytokeratins, structural proteins within cells, have important roles in e.g., apoptosis, mitosis and cell signaling. High levels of ccCK-18 levels have been reported in sepsis patients, and ccCK-18 has been found to have prognostic value also in other critically ill patient groups [70, 71].

We evaluated ccCK-18 in hematological patients with FN. Elevated ccCK-18 level was associated with severe sepsis, ICU admission and death but not with bacteremia [72]. On d1, the area under the curve value was 0.767, which was slightly higher than that for CRP or PCT.

As this is the only ccCK-18 study in hematological patients with FN, further studies are needed. Apoptosis plays an important role regarding the complications of sepsis [73]. The combination of an inflammatory marker and an apoptosis marker, such as ccCK-18, might further improve prognostics.

Other biomarkers evaluated during the HEMATULEHDUS study

We have also evaluated high sensitivity troponin T as well as micro-RNAs (Cohort 2), but these studies have not been published yet. Serum citrulline, alpha1-acid-glycoprotein, intestinal fatty acid binding protein, and zonulin have been analyzed from the most recent cohort 3, and the results will be published soon.

Discussion

This single-center prospective long-term study involved three study cohorts which aimed to identify novel diagnostic and prognostic biomarkers in hematological patients treated for FN at hematological ward after intensive chemotherapy. Based on both national and local epidemiological data, we selected AML patients receiving conventional anthracycline and cytarabine-based chemotherapy and AHCT recipients to be included in our study population. These patients have a high risk of FN ranging from approximately 80% to almost 100%, and a substantial proportion has complicated courses, with some patients also requiring ICU admission.

So far, approximately 300 sepsis biomarkers have been evaluated in clinical studies [74]. They have been mainly studied at emergency departments or ICUs. Only a fraction of these biomarkers has been specifically evaluated in hematological patients with FN. Most study cohorts, including ours, have been small, including fewer than 100 patients. We selected biomarkers for our study series through several stages of literature search. Most biomarkers had been previously evaluated in intensive care units or emergency departments in non-neutropenic patient populations diagnosed with sepsis, but only a few of them had been evaluated in hematological patients with FN. The suitability of biomarker kinetics for clinical purposes and logistical considerations, such as the availability of commercial reagents and diagnostic tests were taken into account. The known association of the biomarker with the pathophysiological background of sepsis was considered an advantage in biomarker selection. Table 2 shows the association of different FN biomarkers with the potential pathophysiological background of sepsis.

Van der Velden's systematic review and meta-analysis included fifty-two studies with nearly 14,000 febrile episodes in over 7,000 immunocompromised children [75]. Six out of the forty-three biomarkers investigated (CRP, PCT, IL-8, IL-6, IL-10, and TNF- α) were significantly associated with bacterial infection at admission. No clear superiority was observed for any biomarker in comparison with PCT (vs. IL-6, IL-8, or IL-10). There is great heterogeneity in the biomarkers studied, as well as in the cut-off values and definitions used, which complicates research.

The mortality of sepsis patients treated at ICU is much higher than that of FN patients treated at the hematology ward. In a Finnish multicenter study involving most ICUs in Finland over a four-month period, ICU mortality of patients with severe sepsis ($n = 470$) was 15.5% and in-hospital mortality 28.3% [76]. The proportion of patients with malignancy was < 5% in that series. In our single center prospective study of AML patients receiving intensive chemotherapy, only seven out of 264 FN patients (2.7%) died during the study period (first treatment cycle). Of our adult

Table 2 Potential novel biomarkers of febrile neutropenia grouped according to postulated pathophysiology of sepsis [78]

Biomarker class	Biomarker
Cytokine or chemokine	Interleukin-10
	Interleukin-18
	Interleukin-1-receptor antagonist
	Interleukin-6
Acute phase protein	Tumor necrosis factor alpha
	C-reactive protein
	Lipopolysaccharide binding protein
	Pentraxin-3
Receptors	Procalcitonin
	Presepsin/Soluble CD14
	Soluble urokinase plasminogen activator receptor
Cellular markers	Soluble CD25
	Soluble HLA-DR
Coagulation-related markers	Antithrombin
	Fibrinogen
	Plasminogen activator inhibitor
	Tissue inhibitor of matrix metalloproteinase-1
Markers of microcirculation	Angiopoietins
	Thrombomodulin
	Vascular endothelial growth factor
Markers of vasodilatation	Adrenomedullin/ pro-adrenomedullin
	Asymmetric dimethylarginine
Markers of organ dysfunction	Brain-type natriuretic peptide
	Endothelin-1
	Micro-RNA-122
Markers of apoptosis	Caspase-cleaved cytokeratin-18
	Cell-free plasma DNA
	FAS ligand (CD95L)/Soluble FAS
Other markers	Citrulline
	Copeptin
	Matrix metalloproteinase-10

AML patients treated in 2007–2018 at Kuopio University Hospital ($n = 125$), 3.5% died due to infectious complications after the first induction course and altogether 9.5% due to septic complications during induction and consolidation courses [77].

Table 1 summarizes the three study cohorts with major endpoints of interest. The patient population of each cohort is quite small, reducing statistical power. The most frequent endpoint assessed in our studies was bacteremia, observed in approximately 20% of AHCT recipients and 50% of AML patients after intensive chemotherapy. Only approximately 5% of patients required ICU admission. Therefore, many markers studied in ICU settings with relatively good predictive power in sepsis patients performed less well in our patient cohorts, in which approximately 95% of patients were treated at the hematology ward.

Severe cytopenias and immunosuppression after chemotherapy are typical for our patients and can sometimes have a significant impact on biomarker production [78]. This applies especially to markers produced by neutrophils, such as suPAR. Leukocyte counts affect cf-DNA levels. This should be kept in mind because usually these patients have leukopenia $< 0.5 \times 10^9/l$, at least at the beginning of FN. Furthermore, VEGF levels are usually low in patients with severe thrombocytopenia. When immunosuppression is significant, it may affect the production of several biomarkers. However, CRP or PCT do not appear to be dependent on cytopenias, so their use is also suitable for cytopenic patients.

The best biomarker for FN would be related to diagnosis, prognosis and/or assessment of treatment response. It should have rapid production/release and a relatively short half-life to be useful in de-escalation strategies. It should not be dependent on blood counts (leukocytes, neutrophils, lymphocytes, platelets) or the underlying hematological disease. The optimal marker should be available 24/7 and suitable for routine laboratory work and automated methods.

Conclusions

Based on our studies in FN hematological patients with more than 20 biomarkers, we suggest further studies of sCD14, IL-1Ra, IL-10, TIMP-1 and ccCK-18 in this patient group. Hopefully, these studies will be multicenter-based and include hundreds of hematological patients to increase statistical power. Metabolomics also needs to be assessed in more detail. Very few novel markers have shown superiority over CRP or PCT [74, 75]. At present, novel biomarkers do not substitute conventional markers but may be additive. They might be best used in conjunction with clinical scoring systems such as qSOFA [4] or NEWS (National Early Warning Sign) [79–81]. Such studies have been conducted for example, on PCT [82, 83] and mid-regional pro-adrenomedullin [84]. Of note, according to Boekhoud et al. [81], qSOFA underestimates the risk of adverse events in patients receiving high-dose glucocorticoids, and the use of NEWS instead of qSOFA is suggested to assess disease severity in immunocompromised patients.

Despite active search, few biomarkers have shown major promise compared to conventional biomarkers and at present cannot replace them. Frequent clinical evaluation is the cornerstone in management of hematological patients with FN, and biomarkers have only a supplemental role here. Using multiple markers simultaneously that assess different aspects of sepsis (e.g. inflammatory response, apoptosis and markers of organ dysfunction) could be beneficial but would complicate the use of decision-making algorithms in

clinical practice. Perhaps the best way to proceed is to use daily sepsis scoring system with 1–2 biomarkers in the early days of FN.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10238-025-01995-3>.

Author contributions Esa Jantunen conceived the idea for this article and prepared the first draft, but all authors contributed to the study design. Sari Hämäläinen, Esa Jantunen, Kari Pulkki and Auni Juutilainen participated in the conduct of the literature review and critically revised the manuscript. All authors commented on each version of the manuscript. All authors have read and approved the final manuscript.

Funding No funding was received for preparing this manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval/Consent The study protocol was accepted by the Ethical Board of North Savo Hospital District in 2006 (100/2006) with several subsequent amendments.

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